

Digit And Benford-Style Tests For Election Forensic Analytics: A Weak-Signal Method For Investigative, Non-Certifying Review

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Abstract

Digit-distribution tests compare reported election counts to simple reference patterns: first-digit Benford frequencies, second-digit frequencies, last-digit uniformity, and related distance or chi-square summaries. This paper places those tests inside the W-series forensic report contract defined by the RCOUNT election-forensics overview. The boundary is deliberately strict. Digit and Benford-style tests can prioritize investigation, but they cannot prove fraud, certify correctness, or replace risk-limiting audits. They are also among the weakest forensic screens for election counts. Precinct totals often have bounded ranges and similar magnitudes, so the assumptions behind Benford’s law are commonly absent; deviations are expected and weakly diagnostic. The method defined here treats digit tests as low-weight, caveated anomaly screens over hash-bound RCOUNT summaries and large count tables, with fixtures that include both planted digit manipulation and innocent failures caused by precinct-size structure.

1 Introduction

Digit-distribution tests ask whether reported election counts have unusual leading, second, or final digits. They are attractive because they are simple: a count table can be reduced to a reference distribution, a test statistic, and a ranked list of contests or jurisdictions that deserve a closer look. In the W-series, that is all they are allowed to do. They can prioritize investigation. They cannot prove fraud, certify correctness, or replace a risk-limiting audit.

This paper also leads with a second boundary specific to this method family. The election-forensics literature is skeptical of Benford’s law for vote counts. Benford behavior depends on measurements that span several orders of magnitude and are not tightly bounded by administrative design. Precinct vote counts often do the opposite: jurisdictions build precincts to be similar in size, contests are bounded by registration and turnout, and many tables cover a narrow range of magnitudes. First-digit tests are therefore fragile, and second-digit tests are especially easy to misread. Deviations from Benford-style references are expected in many innocent election tables.

The purpose of W.04 is to define a cautious method wrapper for these weak screens. Following the W.01 framework contract [Della-Libera, 2026b], it specifies the digit-test families, the RCOUNT inputs they consume, the transcript fields they must emit, and the fixture roles that keep both planted anomalies and innocent failures visible. It is a method and framework paper. No forensics crate, function, fixture, or test is claimed to exist here.

2 Method

A digit-test report begins with a table of nonnegative reported counts and a chosen unit of analysis: contest, jurisdiction, reporting unit, batch, or a larger count table. The method extracts a digit feature and compares its observed distribution to a stated reference distribution. The comparison may be summarized by a chi-square statistic, an absolute or squared distance, a ranked-residual score, or another predeclared distance measure. The statistic is an anomaly score, not a verdict.

Family	Reference and caveat
First digit	Compare leading digits 1 through 9 with the Benford reference, $P(d) = \log_{10}(1 + 1/d)$. Use only where the count table plausibly spans broad magnitudes; many precinct tables do not.
Second digit	Compare digits 0 through 9 with the corresponding Benford-style second-digit reference. Treat results as especially weak for vote counts, because modest range restriction and precinct design can move the distribution.
Last digit	Compare terminal digits 0 through 9 with a uniform reference, or with a pre-declared administrative baseline if rounding or reporting rules affect final digits. Last-digit concentration can be a screen, but it can also reflect aggregation, batch construction, or data-entry conventions.

For an observed count O_d and expected count E_d in digit bin d , a common summary is

$$X^2 = \sum_d \frac{(O_d - E_d)^2}{E_d},$$

provided the expected counts are large enough for the approximation to be meaningful. A simpler distance score such as $\sum_d |O_d - E_d|$ may be reported instead when the goal is ranking rather than calibration. If a p-value is reported, the transcript must state the reference distribution, exclusions, and baseline population that make the calibration meaningful.

The central diagnostic warning is part of the method. Vote counts often violate Benford assumptions for innocent reasons: bounded electorates, similar precinct sizes, contest-specific turnout ceilings, truncation by reporting-unit design, and pooling choices that mix unlike jurisdictions. A high statistic may indicate a data feature worth explaining, but it is weak evidence by itself and must be read with the structure of the count table.

3 RCOUNT Inputs And Report Contract

RCOUNT can feed this method with normalized summaries and large count tables: contest totals by reporting unit, jurisdiction, candidate, batch, or other publicly reproducible

grouping. The method must bind every report to the source package hashes so that a second analyst using the same RCOUNT package and the same registered method recomputes the same extracted digits and scores [Della-Libera, 2026a].

W.04 instantiates the W.01 transcript fields as follows.

Field	W.04 instantiation
<code>method_id</code>	Registered digit-distribution screen.
<code>feature_set</code>	Count column, digit position, exclusions, pooling rule, and any minimum-count threshold.
<code>baseline_population</code>	Contest, jurisdiction set, election, time window, and comparable count table used for the reference.
<code>model_id</code>	<code>digit-test/benford</code> , with subtype for first digit, second digit, last digit, or declared distance family.
<code>unit_id</code>	Contest, jurisdiction, reporting unit, batch, or aggregate count table being scored.
<code>score</code>	Test statistic or distance from the declared reference.
<code>p_value_or_rank</code>	Optional p-value, percentile, or queue rank, reported only with its assumptions.
<code>investigation_note</code>	Required weak-signal caveat: digit tests are low-weight screens, deviations are expected under bounded or similar precinct sizes, and the result is not evidence of fraud.

The source hashes, feature definition, baseline population, and model identity are part of the evidence. Changing any of them changes the question. The investigation note is not optional boilerplate; it is the field that prevents a large digit statistic from being circulated as a certification or fraud claim.

4 Fixture Pattern

The intended fixture family follows the four-role W-series pattern. These are defined roles for later implementation, not a claim that fixture files or tests already exist.

Fixture	Role
No-anomaly	A synthetic count table drawn from a plausible election-count distribution with no planted digit manipulation. A cautious method should return low priority or explain ordinary sampling variation.
Injected tampered digit	A table with a planted digit-pattern distortion, such as excess repeated final digits or altered leading digits in a subset. The method should surface the affected unit while still labeling the result as investigative.
Innocent digit deviation	A bounded precinct-size table that deviates from a Benford-style reference for structural reasons. This exercises the false-positive warning and should not be described as suspicious by itself.
Boundary failure	A table that formally “fails” a digit test but is fully explained by precinct-size design, turnout ceilings, or pooling choices. The report must carry the explanation in <code>investigation_note</code> .

The last two fixtures are as important as the planted one. A digit method that only finds the injected distortion is incomplete; it must also demonstrate that ordinary administrative structure can produce high statistics and that the report language keeps those statistics in their proper, weak-signal role.

5 Boundaries

Digit and Benford-style tests are weak forensic screens. They are easy to run, easy to rank, and easy to misread. A high score can justify asking why a count table has a particular digit pattern; it cannot answer that question by itself. It is not proof of fraud, not a certification failure, and not a substitute for risk-limiting audits or ordinary canvass reconciliation.

The main technical boundary is the Benford assumption. Many election-count tables do not span the orders of magnitude needed for leading-digit behavior. Precincts are often intentionally similar in size; contests impose upper bounds; turnout, registration, and reporting-unit rules shape the available numbers. Second-digit tests add another layer of fragility and should be treated with particular caution. Last-digit tests avoid the Benford premise but remain sensitive to aggregation, rounding, transcription, and reporting conventions.

Accordingly, W.04 reports should never be used alone. They should be a low-weight signal in a broader investigation queue, read alongside source hashes, lineage, audit records, batch information, geography, and human review. The lead boundary from W.01 remains controlling: W-series outputs prioritize investigation; they do not prove fraud, certify correctness, or replace audits.

6 Conclusion

Digit-distribution tests can be part of an election-forensics workbench, but only in a narrow and carefully labeled role. This paper defines that role for RCOUNT packages: extract stated digit features from hash-bound summaries or large count tables, compare them to declared references, report a statistic or rank, and attach the weak-signal caveat required by the W.01 contract.

The strongest contribution is the boundary, not the statistic. Because election counts often violate Benford assumptions for innocent administrative reasons, first-digit, second-digit, and last-digit tests should be treated as low-weight screens. They may help decide where to look next. They do not establish what happened, and they must never be presented as fraud evidence or certification evidence on their own.

References

Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.

Gio Della-Libera. Election forensic analytics for rcount packages. Technical report, Apportionment Research, 2026b.